

ENHANCING TOURISM DIGITAL TWINS WITH REAL TIME ,BILATERAL DATA ACROSS SRI LANKA

PROJECT ID : 25-26J-380



MEET OUR TEAM



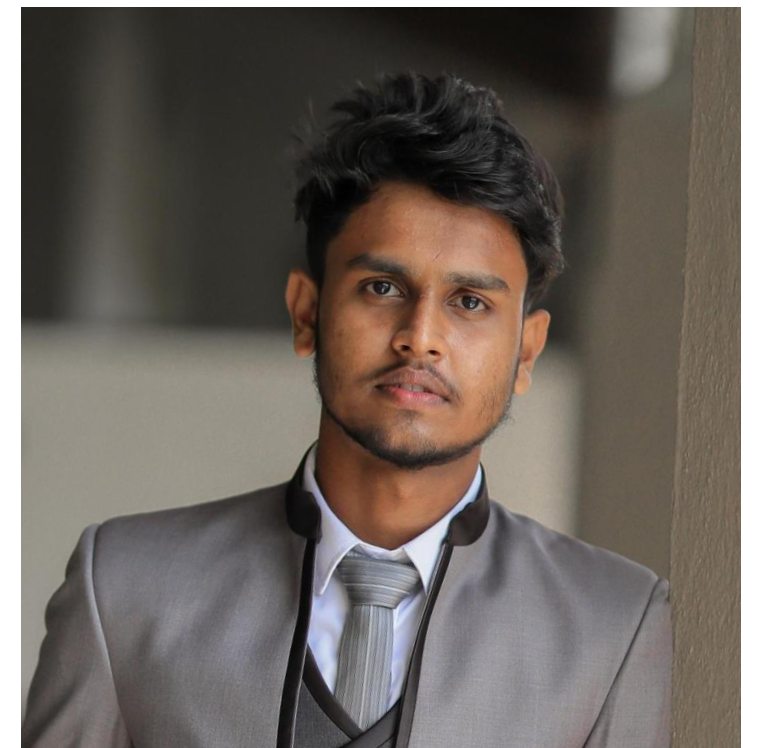
**GROUP
LEADER**
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DINUSHA



ENHANCING TOURISM DIGITAL TWINS WITH REA

INDIVIDUAL COMPONENTS



REAL-TIME MONITORING & VISUALIZATION MODULE

Specialization: Information Technology

Real Time Monitoring And Visualization Module

Problem Definition

- Existing Digital Twins are static
- No immersive way to experience tourist destinations remotely.

Creative Solution

- Develop a real-time, dynamic digital twin that updates live weather and environmental changes according to the time.

Real Time Monitoring And Visualization Module

Research

Gap

Research Paper	Real-Time weather simulation	Time-of the day/lighting simulation	sensor-based environmental updates	Immersive 3D Experience
Florido-Benitez (2024) - DTs for smart destinations [1]				
Boboc et al. (2022) - AR in cultural heritage [2]				
Verde et al. (2023) - BLE beacons in museums [3]				
Singh et al. (2023) - Map-matching review [4]				
Ruiz-Meza & Montoya-Torres (2022) - TTDP SLR [5]				
Proposed System				

Real Time Monitoring And Visualization Module

Key Domains

- Computer Graphics and 3D Modeling
- Real-Time Data Integration
- Human-Computer Interaction

Latest Technologies



Real Time Monitoring And Visualization Module

Evaluation Plan

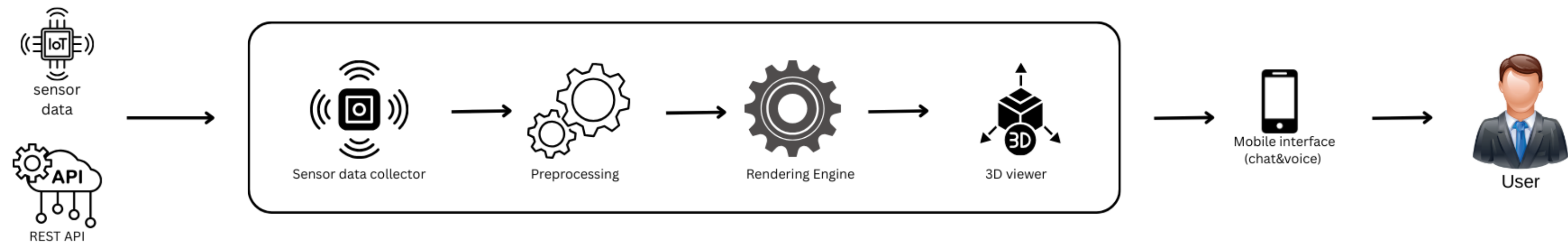
- Accuracy Testing
- Performance Testing
- Usability Testing
- Comparative Analysis

Data Availability & Ethical Clearance

- Data Sources: OpenWeatherMap, global time APIs
- Ethical Assurance
- Cultural sensitivity
- Compliance

Real Time Monitoring And Visualization Module

System Overview



Real Time Monitoring And Visualization Module

Real world

Scenario

- Displays a realistic 3D/360° model of Sigiriya's rock fortress and surroundings.
- Simulates **real-time weather changes** like rain, sunshine, fog, and clouds.
- Shows **dynamic lighting effects** reflecting actual time of day (morning, noon, evening, night).
- Reflects **environmental conditions live**, making the virtual site feel “alive”.
- Allows users to **navigate and explore** Sigiriya virtually from different angles and distances.



LOCATION-BASED DETAILS GATHERING & MANAGEMENT

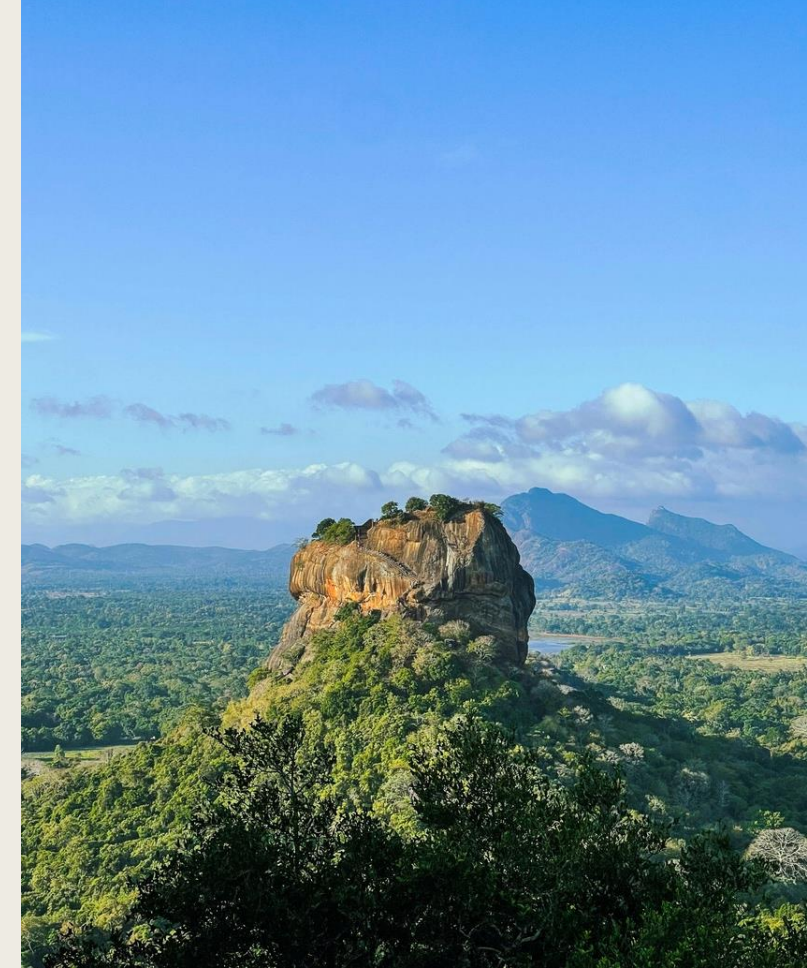
Specialization: Information Technology

LOCATION-BASED DETAILS GATHERING & MANAGEMENT

Problem

Definition

- Existing tourism apps don't provide real-time, on-site guidance.
- GPS-only tracking is unreliable in forests, caves, and hilly areas.
- Current apps suggest routes based on distance, not walking time.
- Tourists get a generic experience, not personalized recommendations.
- Current platforms lack real-time visitor-side intelligence.



Solution

- [Detects tourist presence at landmarks & sub-landmarks.](#)
- [Delivers instant, contextual content upon arrival.](#)
- [Uses Next Nearest Pinpoint Identification \(NNPI\) for time-aware routing.](#)

LOCATION-BASED DETAILS GATHERING & MANAGEMENT

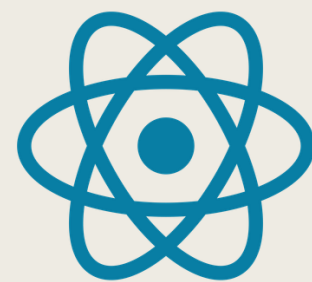
Research Paper	On-Site Unlock (Geo-lock / POL)	Sub-Landmark Arrival Detection	Time-Aware Routing (NNPI)
Florido-Benitez (2024) - DTs for smart destinations [1]			
Boboc et al. (2022) - AR in cultural heritage [2]			
Verde et al. (2023) - BLE beacons in museums [3]			
Singh et al. (2023) - Map-matching review [4]			
Ruiz-Meza & Montoya-Torres (2022) - TTDP SLR [5]			
Proposed System			

LOCATION-BASED DETAILS GATHERING & MANAGEMENT

Key Domains

- Smart tourism & cultural-heritage informatics
- Geospatial & positioning
- Artificial Intelligence & Machine Learning
- Knowledge engineering & content authoring

Technologies



LOCATION-BASED DETAILS GATHERING & MANAGEMENT

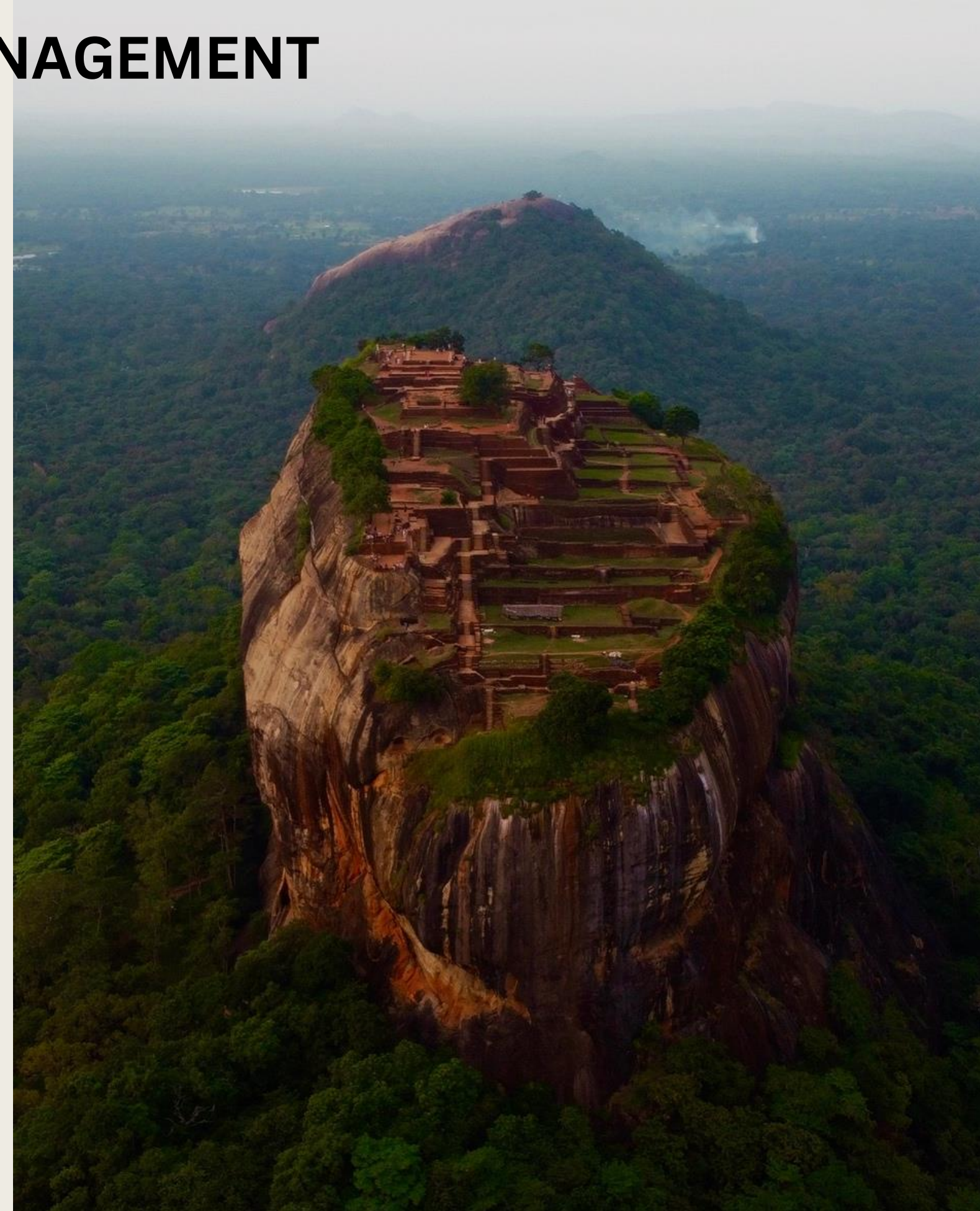
Evaluation

Plan

- Arrival Detection Accuracy
- Next Nearest Pinpoint Identification (NNPI) Performance
- Adaptive Tour-Wide Time Estimation
- On-Site Content Unlock Reliability
- System Performance
- Generalizability & Robustness

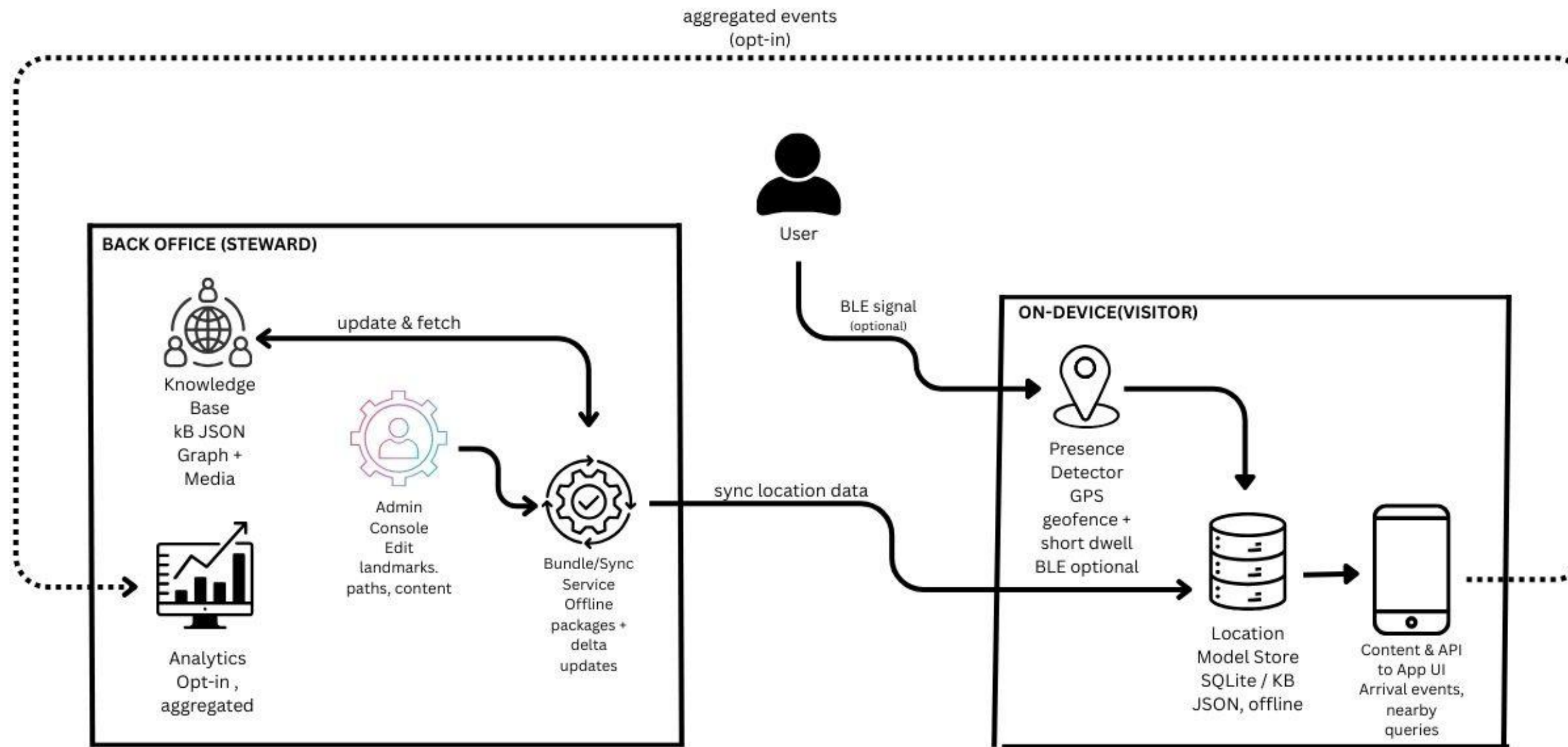
Data Availability and Ethical Clearance

- Personally visiting heritage locations to gather accurate data.
- Collecting real-time visitor location data using GPS Tracking



LOCATION-BASED DETAILS GATHERING & MANAGEMENT

system overview diagram



LOCATION-BASED DETAILS GATHERING & MANAGEMENT

Reference

- [1] L. Florido-Benítez, “The Use of Digital Twins to Address Smart Tourist Destinations’ Future Challenges,” *Platforms*, vol. 2, no. 4, pp. 234–254, Dec. 2024, doi: 10.3390/platforms2040016. [MDPI](#)
- [2] R. G. Boboc, E. Băutu, F. Gîrbacia, N. Popovici, and D.-M. Popovici, “Augmented Reality in Cultural Heritage: An Overview of the Last Decade of Applications,” *Applied Sciences*, vol. 12, no. 19, 9859, 2022, doi: 10.3390/app12199859. [MDPI](#)
- [3] D. Verde, J. Romero-González, J. L. López-Sánchez, and A. Brandão, “Indoor Content Delivery Solution for a Museum Based on BLE Beacons,” *Sensors*, vol. 23, no. 17, 7403, 2023, doi: 10.3390/s23177403. [MDPI](#)
- [4] S. Singh, J. Singh, S. B. Goyal, M. El Barachi, and M. Kumar, “Analytical Review of Map-Matching Algorithms: Analyzing the Performance and Efficiency,” *Archives of Computational Methods in Engineering*, vol. 30, pp. 4897–4916, 2023, doi: 10.1007/s11831-023-09962-5. [SpringerLink](#)
- [5] J. L. Ruiz-Meza and J. R. Montoya-Torres, “A Systematic Literature Review for the Tourist Trip Design Problem: Extensions, Solution Techniques and Future Research Lines,” *Operations Research Perspectives*, vol. 9, 100228, 2022, doi: 10.1016/j.orp.2022.100228. [IDEAS/RePEc](#)



INTERACTIVE NLP & PREDICTIVE OPTIMIZATION MODULE

Specialization: Information Technology

INTERACTIVE NLP & PREDICTIVE OPTIMIZATION MODULE

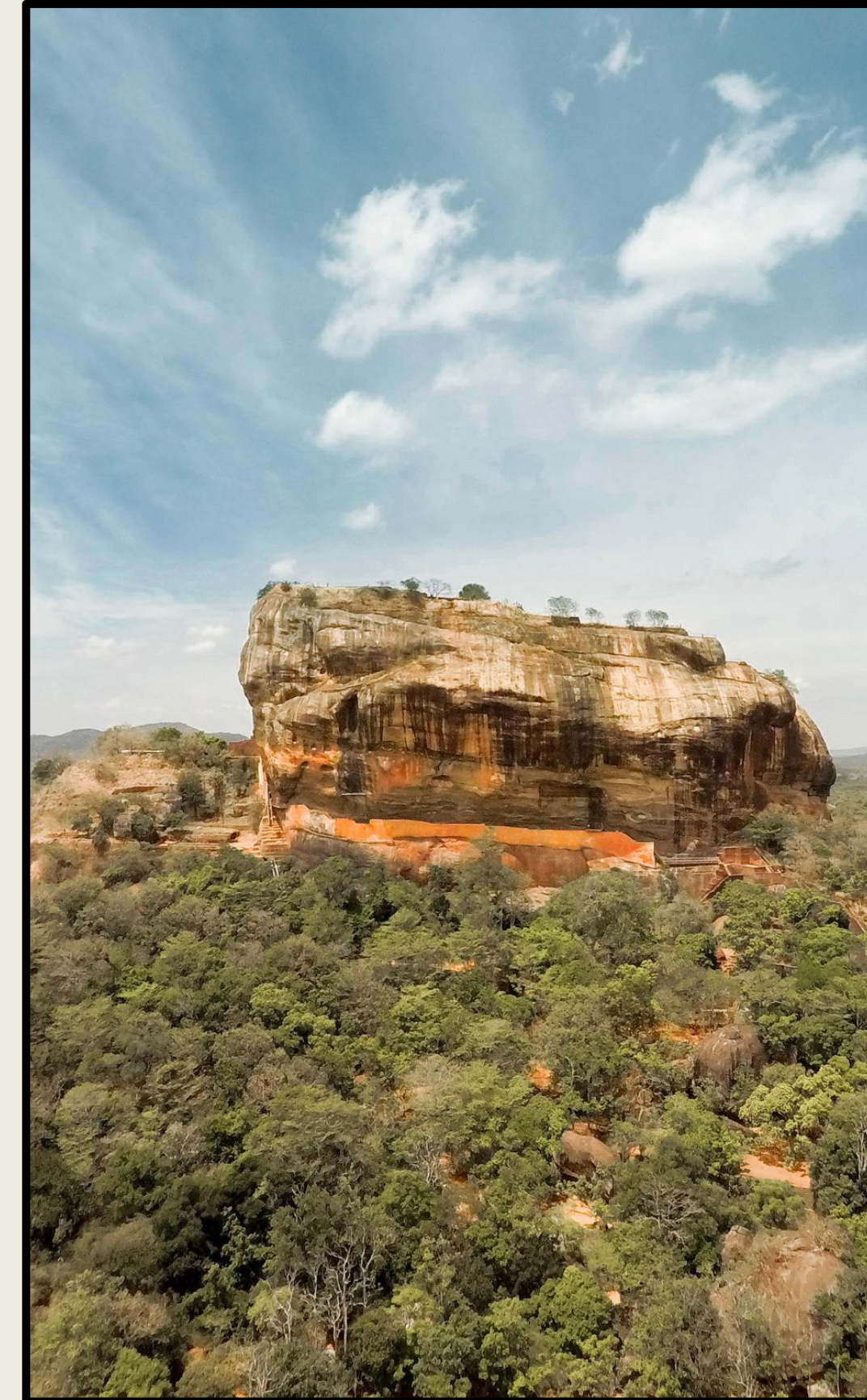
Problem

Definition

- Current tourism digital twins only show data, not actionable insights.
- Tourists struggle to interpret raw sensor numbers.
- Chatbots in tourism are FAQ-based, not linked with live digital twin data.
- No system gives future predictions + alternative site recommendations.
- Visitor feedback is not used to update systems in real-time.

Solution

- Interactive NLP assistant → converts raw sensor data into clear, human-friendly answers
- Context-aware recommendations → suggests alternative sites or best visiting times when the main location is overcrowded.
- Predictive optimization → uses historical + live data to forecast future conditions
- Two-way interaction → tourists and managers can ask questions in natural language via chat/voice.
- Feedback loop integration → system adapts based on real visitor feedback, updating alerts and manager actions in real-time.



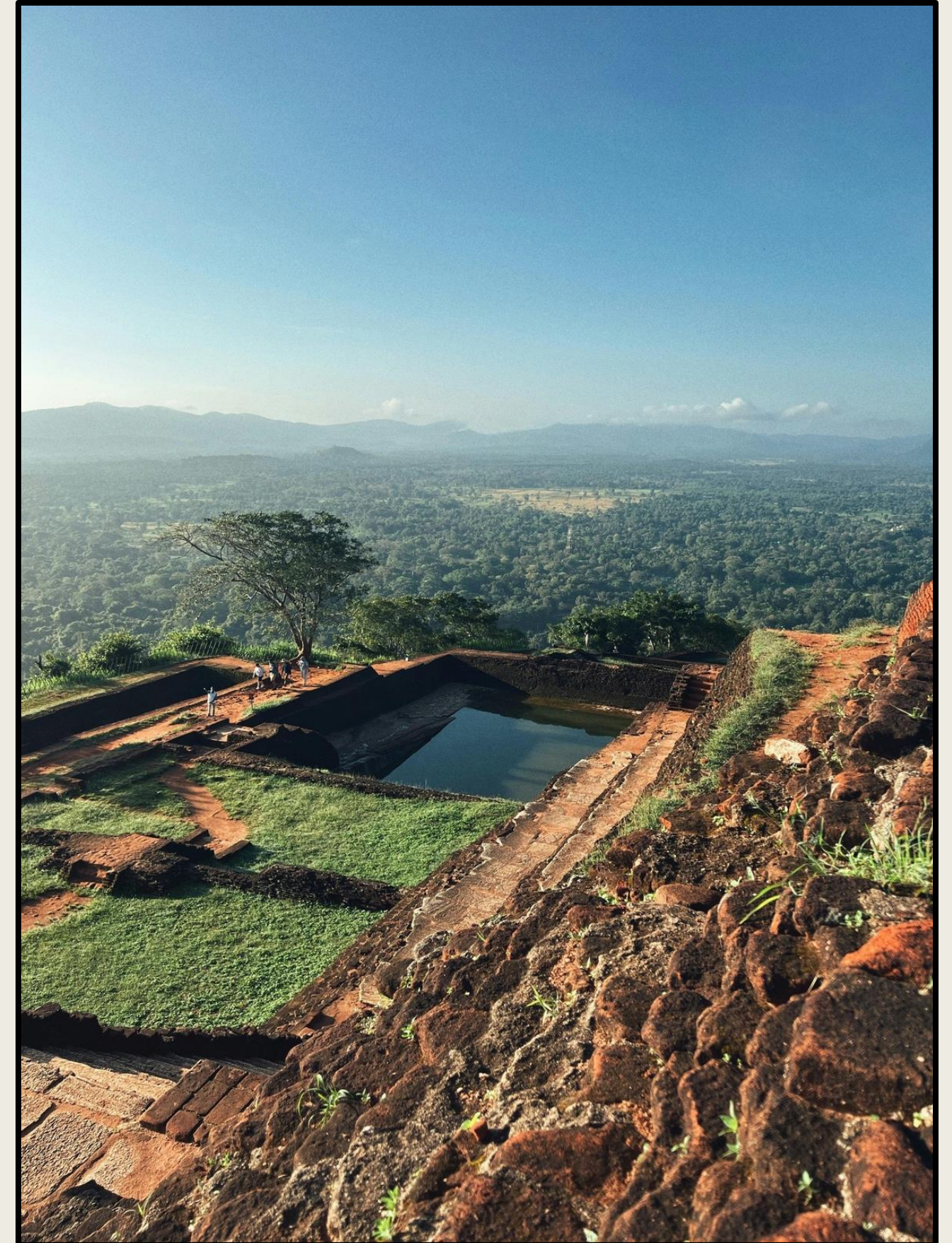
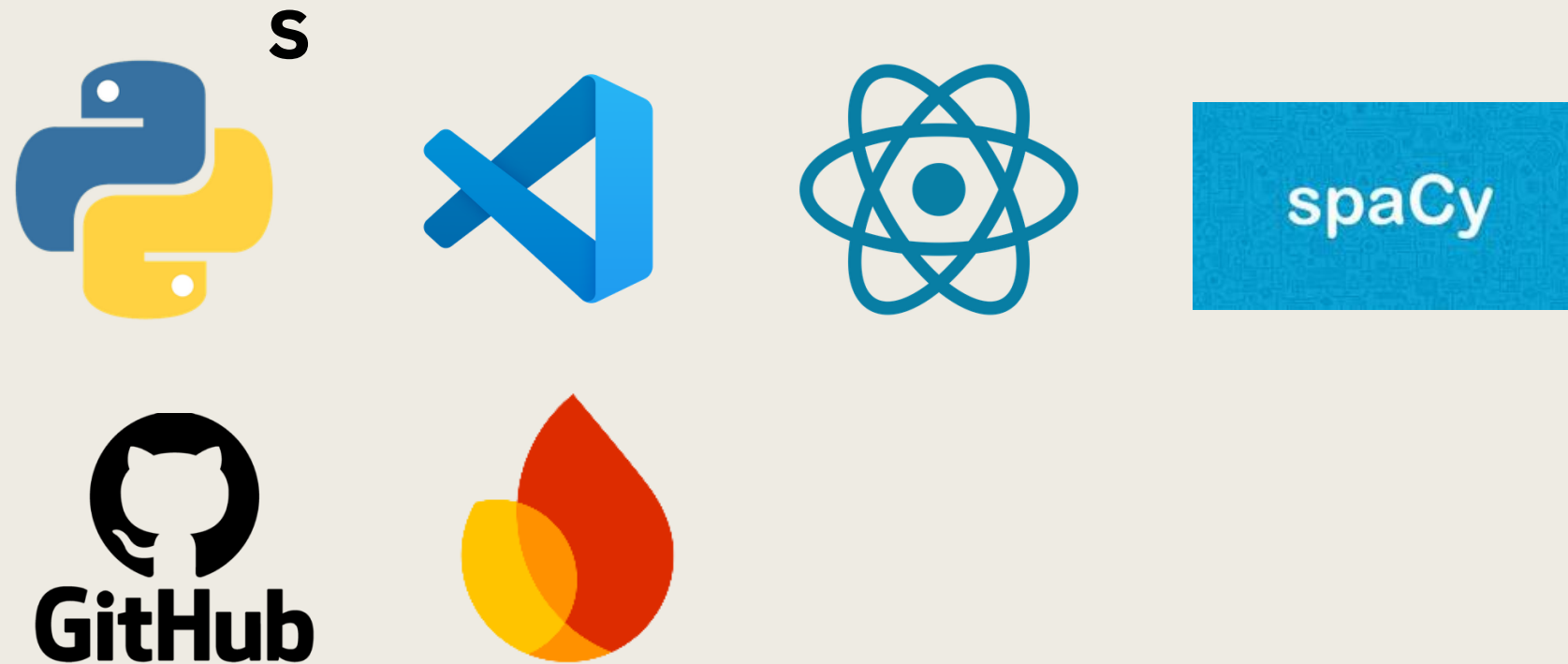
INTERACTIVE NLP & PREDICTIVE OPTIMIZATION MODULE

Key Domains

Used

- Natural Language Processing (NLP)
- Predictive Analytics & Machine Learning
- Recommendation Systems
- Cloud/Web Technologies

Technologies



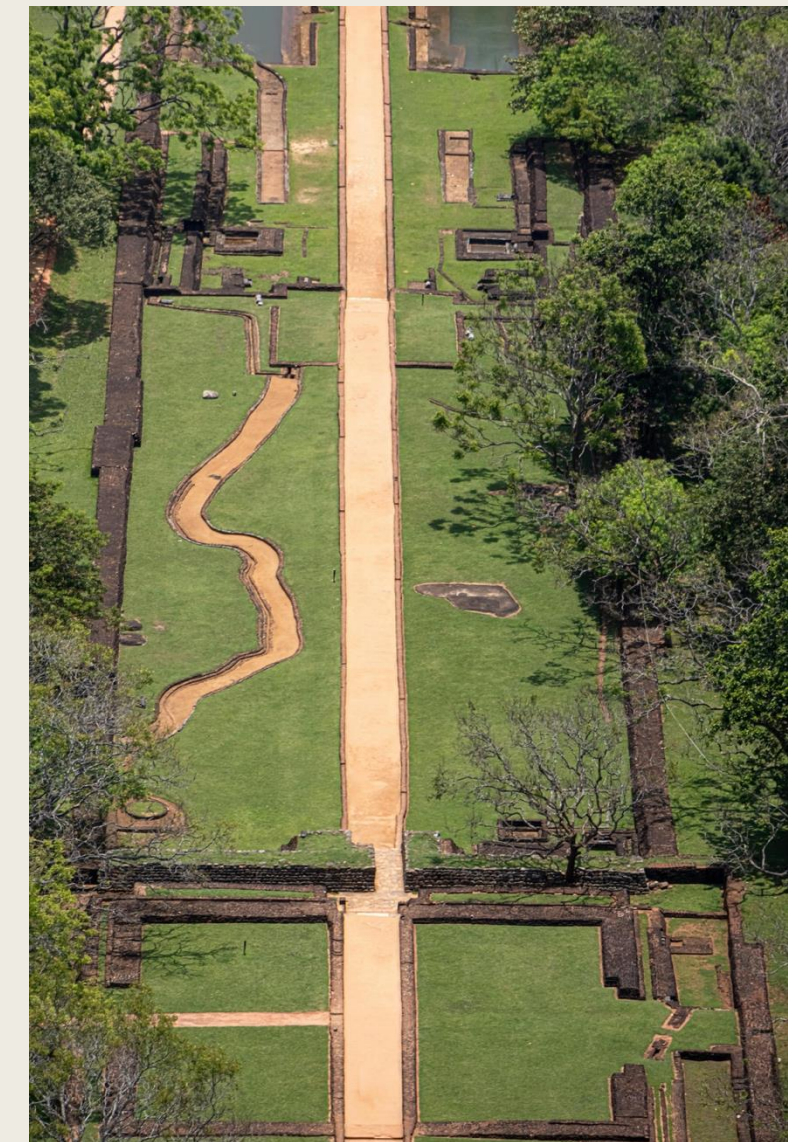
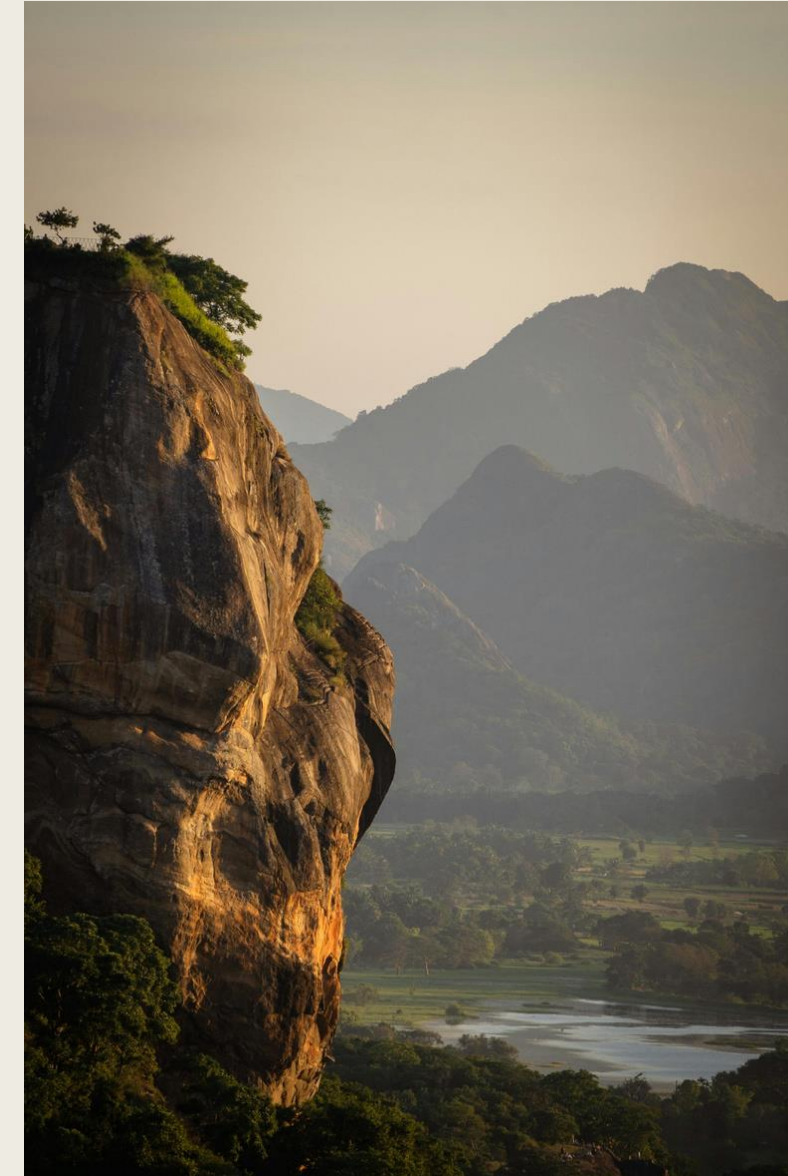
INTERACTIVE NLP & PREDICTIVE OPTIMIZATION MODULE

Evaluation

- Accuracy of predictions
- User satisfaction from pilot testing with tourists & managers.
- Response time (< 2 seconds for queries).

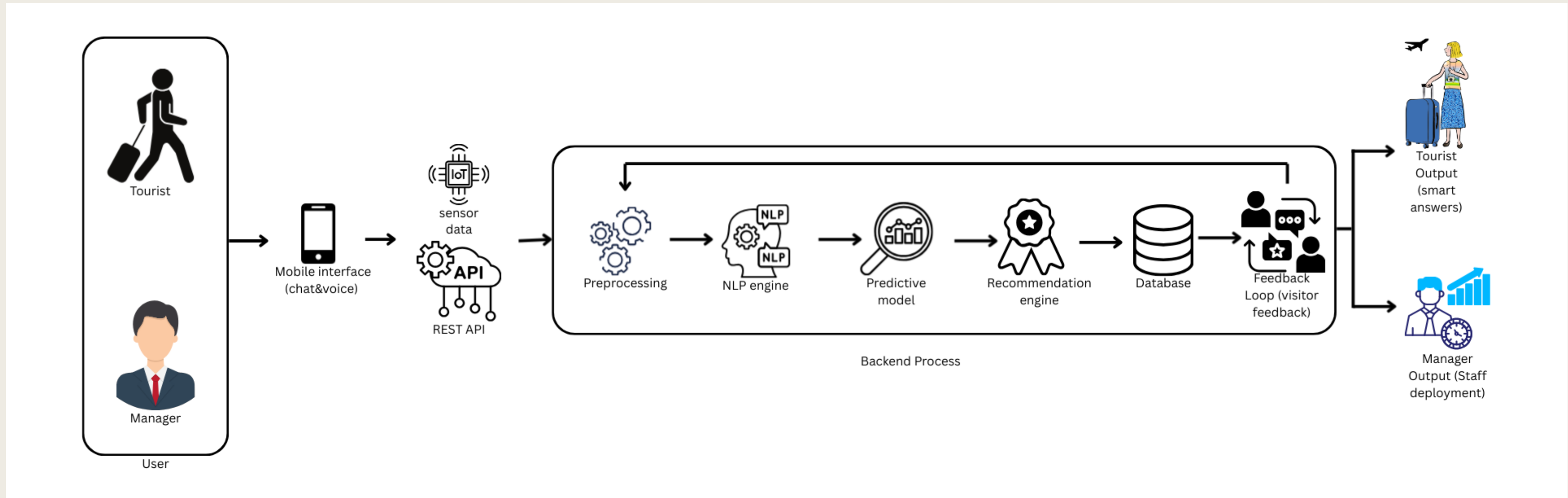
Data Availability & Ethics

- Real-time sensor streams from the digital twin.
- Anonymous tourist queries/feedback (no personal data).
- Weather API + crowd data collection.



INTERACTIVE NLP & PREDICTIVE OPTIMIZATION MODULE

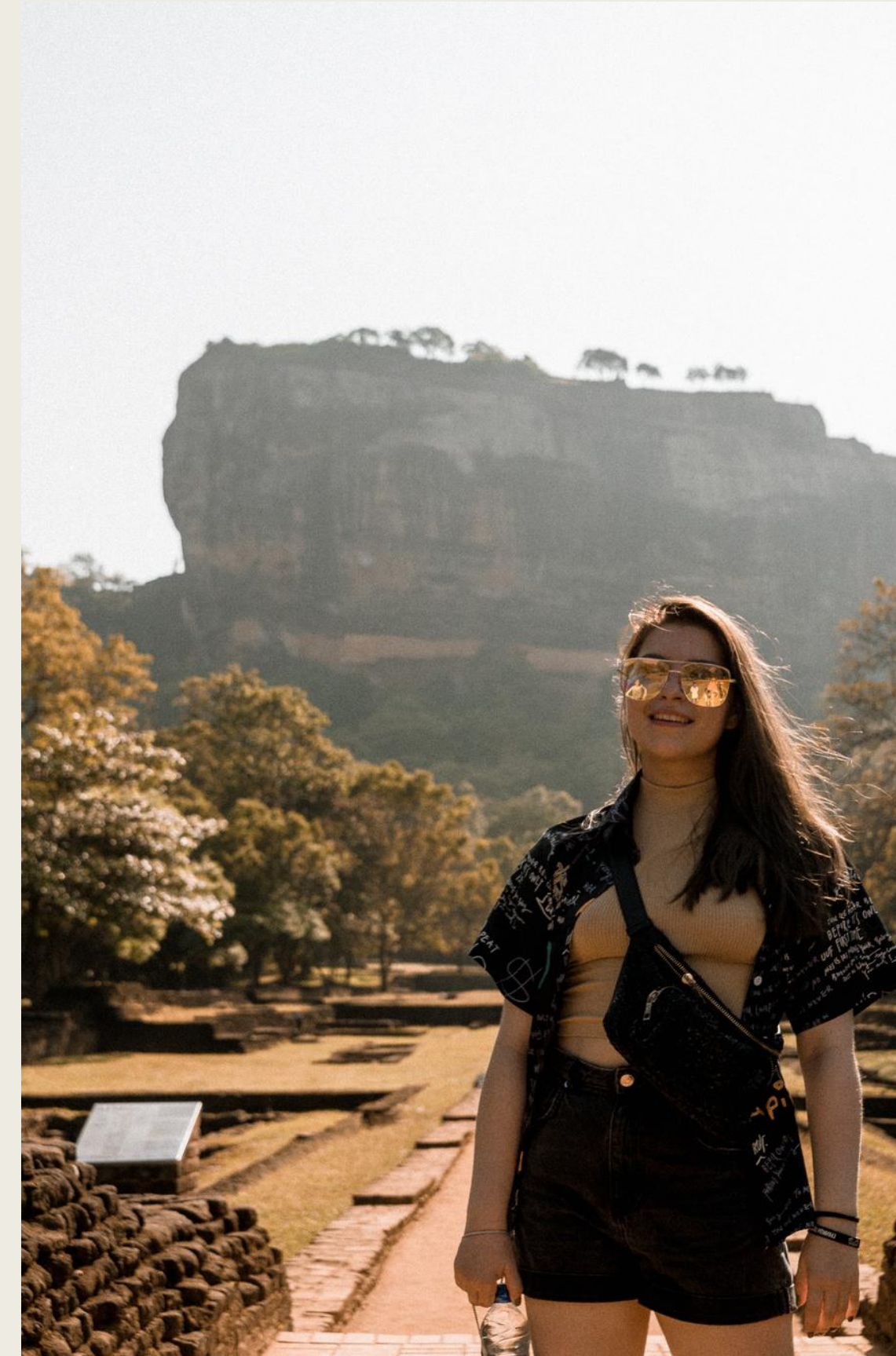
System Overview Diagram



INTERACTIVE NLP & PREDICTIVE OPTIMIZATION MODULE

Real-World Use Case

- Tourist asks: “Is the Sigiriya summit crowded now?” → System: “Yes, waiting time is 25 minutes. Consider visiting Pidurangala Rock nearby.”
- Manager asks: “Where should I send more staff?” → System: “East gate congestion increasing. Deploy 2 guides to manage flow.”



Research Gap

Feature	Zhang et al. (2021) [1]	Chen et al. (2021) [2]	Kim et al. (2021) [3]	Wang & Liu (2020) [4]	odriguez et al. (2019) [6]	Proposed System
Conversational Interface	✗	✗	✓	✗	✗	✓
Real-Time Prediction	✓	✗	✗	✗	✓	✓
Adaptive Learning	✗	✗	✗	✓	✗	✓
Multi-stakeholder Support	✗	✗	✗	✗	✗	✓
Environmental Integration	✓	✗	✗	✗	✗	✓

INTERACTIVE NLP & PREDICTIVE OPTIMIZATION MODULE

Reference

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1. Almeida, D. S. de, Brito e Abreu, F., & Boavida-Portugal, I. (2025). Digital twins in tourism: a systematic literature review — Highlights that most tourism digital twins are one-directional, establishing the need for bilateral interaction.
2. Yi, S., Chen, X., & Tang, C. (2021). Tsformer: Time series Transformer for tourism demand forecasting — Demonstrates advanced predictive modeling techniques in tourism forecasting using transformers.
3. Lemmel, J., Babaiee, Z., Kleinlehner, M., et al. (2022). Deep-Learning vs Regression: Prediction of Tourism Flow with Limited Data — Shows that deep-learning models outperform ARIMA in tourism flow prediction— relevant to your predictive layer.
4. Navamani & Kannammal (as cited in Núñez, 2024). Machine learning applied to tourism: A systematic review — Covers ML and AI's growing role in smart tourism, including chatbots and recommendation systems.
5. "Natural language processing applied to tourism research: A systematic review" — Offers a comprehensive overview of NLP approaches used in tourism—foundation for your NLP interaction layer.
6. Litavniece, L., Kodors, S., et al. (2023). Digital twin: an approach to enhancing tourism competitiveness — Demonstrates how digital twins combined with recommendation systems can boost tourism sector performance.



DESTINATION ANALYTICS AND DYNAMIC CROWD MANAGEMENT SYSTEM

Specialization: Information Technology

DESTINATION ANALYTICS AND DYNAMIC CROWD MANAGEMENT SYSTEM

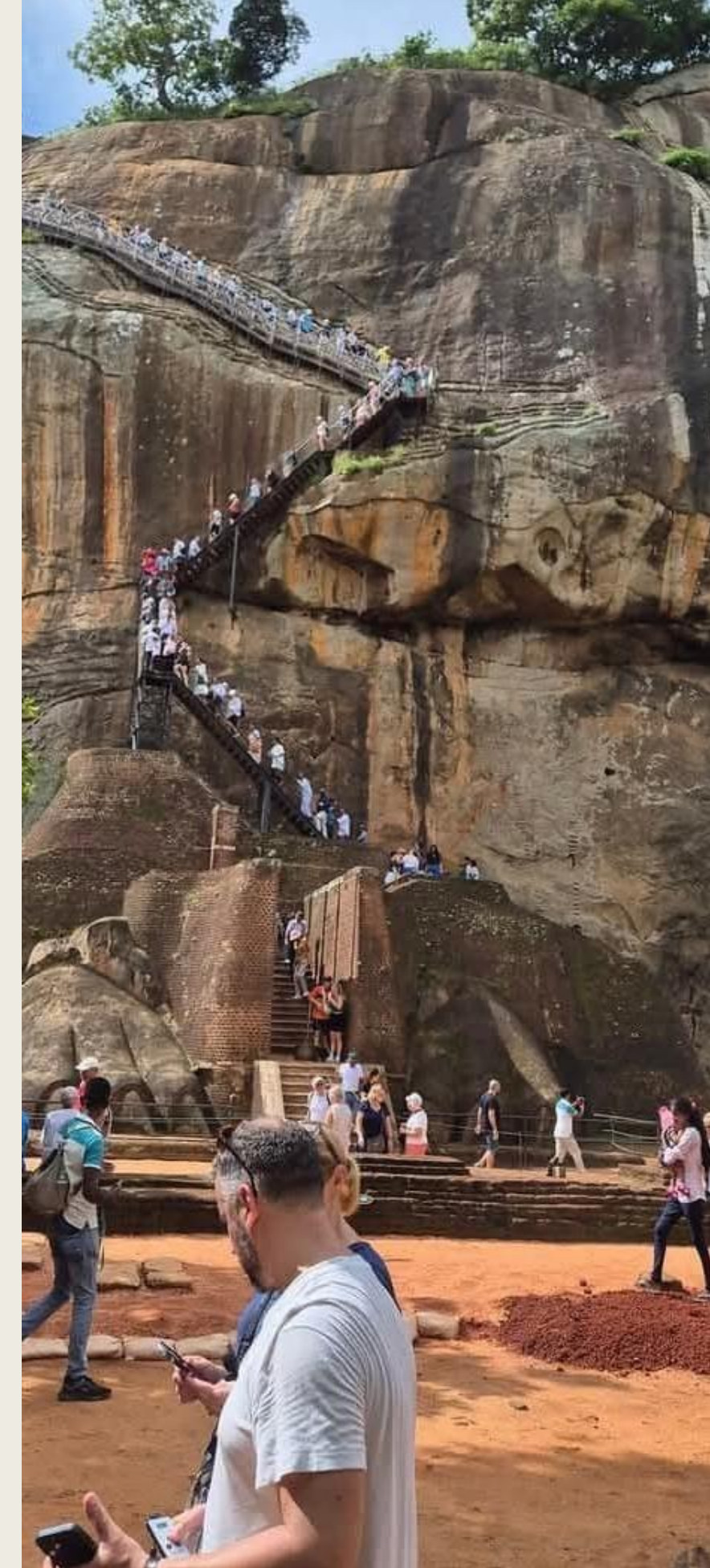
Problem

Definition

- Overcrowding and Visitor Flow Issues
- Lack of Destination-Level Analytics
- Poor Understanding of Visitor Satisfaction in tourist destinations.
- Ineffective Marketing Strategies

Solution

- Crowd Detection (IoT + Computer Vision)
- Marketing Strategy Engine.

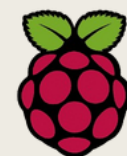
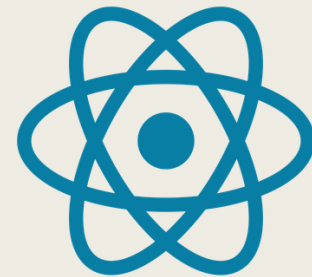


DESTINATION ANALYTICS AND DYNAMIC CROWD MANAGEMENT SYSTEM

Key Domains

- Tourism Management & Marketing
- Data Analytics & Forecasting
- Natural Language Processing
- IoT & Computer Vision
- Decision Support Systems

Technologies



RaspberryPi



DESTINATION ANALYTICS AND DYNAMIC CROWD MANAGAEMENT SYSTEM

Research

Research Paper	Crowd detection	Sentiment Aware Analysis	Tourist Arrival Analysis	Marketing Strategy Engine
Palomeque-González et al. (2025) – IoT & Edge AI for Cultural Heritage Monitoring				
Almeida et al. (2025) – Digital Twins in Tourism: A Systematic Literature Review				
Fronzetti Colladon et al. (2021) – Social Network & Semantic Analysis for Tourism Forecasting				
Singh et al. (2023) – Map-matching review [4]Munar et al. (2021) – Sentiment Analysis for Destination Sustainability				
Ribeiro et al. (2022) – Tourism Destination Management with Sentiment & Geo-location Data				
Proposed System				

DESTINATION ANALYTICS AND DYNAMIC CROWD MANAGEMENT SYSTEM

Evaluation

Plan

- Crowd Detection Accuracy
- Tourist Arrival Forecasting Performance
- Visitor Sentiment Analysis Reliability
- Marketing Strategy Engine Effectiveness
- System Performance
- System Robustness & Integration

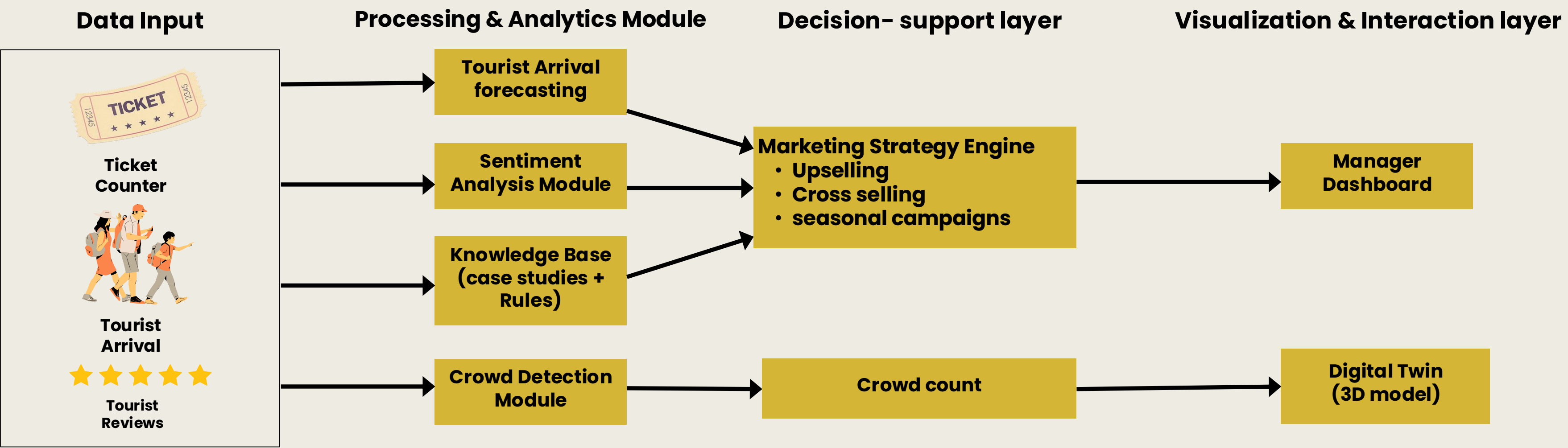
Data Availability and Ethical Clearance

- Ticketing Data
- Visitor Reviews
- IoT Sensor Data
- Expert Validation
- Ethical Compliance



DESTINATION ANALYTICS AND DYNAMIC CROWD MANAGEMENT SYSTEM

System overview diagram



DESTINATION ANALYTICS AND DYNAMIC CROWD MANAGEMENT SYSTEM

Real-World Use

Case

- Location- Sigiriya, Sri Lanka
- forecast says Indians arrive increase in July.
- Sentiment says Indians complain about overcrowding at Sigiriya
- Knowledge base says Thailand solved the overcrowding issue with timed entry campaigns.
- So the output strategy is launch a July campaign for Indians with timed entry tickets. Also do a cross selling by promoting Kandy cultural festival to spread demand.



Dynamic crowd management -

- Location- Sigiriya
- lot sensors are placed at crowd gathering places like entrance, stairs, mirror wall, frescoes etc. Crowd count is gathered and displayed in the 3D model.

**Thank
You**